



Bahria University
Department of Computer Science

Student Result Processing System

Submitted to: Sir Qazi Mohsin Ijaz
Course: Computer Organization & Assembly Language Lab
Course Code: CEL 323
Semester: Spring 2026
Class: BS-CS 4-B (Morning)
Deadline: 17th May 2026

Group Members

Sr.	Name	Enrollment No.
1	Muhammad Sadiq Aslam Khan	01-134242-091
2	Malik Junaid Ahmed	01-134242-061
3	Bilawal Khokhar	01-134242-036

May 2026

Contents

1	Introduction	2
2	Problem Statement	2
3	What We Built	2
4	How We Stored Data	3
5	Assembly Concepts Used	3
6	Program Output Screenshots	4
6.1	Main Menu	4
6.2	Option 1 — Add Student	4
6.3	Option 2 — Show All Records	5
6.4	Option 3 — Search by Roll Number	5
6.5	Option 4 — Sort by Marks	6
6.6	Option 5 — Result Summary	6
7	GitHub Repository	7

Introduction

We built this project for our Computer Organization and Assembly Language Lab course. The idea was simple — the department keeps student records manually, which causes mistakes and takes time. We wrote a menu-driven console program in x86 MASM Assembly to handle that. It lets you add students, view their records, search by roll number, sort by marks, and see a result summary — all from a simple text menu.

We used the Irvine32 library for input/output, and Visual Studio 2022 as our IDE. The full source code is on GitHub:

<https://github.com/JunaidAhmed13/StudentResultSystem>

You can download `main.asm` directly from there.

Problem Statement

Manual record handling is slow. Finding a student means scanning through the whole register. Calculating totals for every student by hand introduces errors. There is no easy way to sort the class by performance or update a record without rewriting things.

We needed a program that stores up to 5 student records, calculates totals on its own, and lets us search, sort, and print results without any manual calculation.

What We Built

The program runs as a console application with 6 options:

Option	Function
1	Add a student (name, roll number, marks for 5 subjects)
2	Show all records with totals, percentages, and pass/fail
3	Search a student by roll number
4	Sort all students by total marks (descending)
5	Show a compact result summary
6	Exit

Pass criteria we set: percentage $\geq 40\%$ is PASS, anything below is FAIL.

How We Stored Data

Everything is in the `.data` segment. We used flat arrays to simulate 2D storage since assembly does not have built-in 2D arrays.

Variable	Type	What it holds
names	db	100 bytes total — 5 students, 20 characters each
rolls	dd	5 dwords, one roll number per student
marks	dd	25 dwords — 5 students times 5 subjects
totals	dd	5 dwords, auto-calculated when a student is added
cnt	dd	Tracks how many students have been entered

To get the name for student i we used `names + i*20`. For marks of subject j of student i we used `marks[(i*5 + j)*4]`. We kept everything as `dd` (32-bit) to avoid register size mismatch errors.

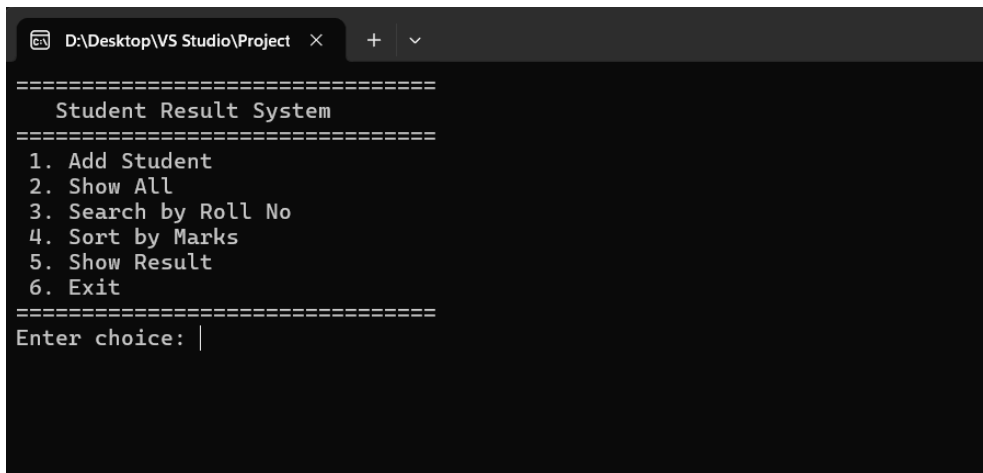
Assembly Concepts Used

Concept	Where we used it
Registers (eax–edi)	Arithmetic, array indexing, I/O calls throughout
<code>.data</code> segment	All strings, arrays, and the counter
Unconditional jump (<code>jmp</code>)	Menu loop, mark input loop, show loop
Conditional jumps	<code>j1</code> , <code>jge</code> , <code>je</code> , <code>jnz</code> for search, sort, pass/fail
Flags (ZF, SF)	Set by <code>cmp</code> before every conditional jump
Counter-controlled loop	<code>markLoop</code> runs 5 times, sort inner/outer loops
Base-index addressing	<code>[edx + esi*4]</code> for all array reads and writes
<code>imul</code>	Index calculations like <code>cnt*5</code> and <code>esi*20</code>
<code>idiv + cdq</code>	Percentage: total divided by 5
Stack (push/pop)	Saving <code>ecx</code> across the inner sort loop
Procedures	<code>pln</code> , <code>addStu</code> , <code>showAll</code> , <code>srch</code> , <code>sortM</code> , <code>showRes</code>
Irvine32	<code>ReadString</code> , <code>ReadInt</code> , <code>WriteDec</code> , <code>WriteChar</code> , <code>Crlf</code> , <code>Clrscr</code>

Program Output Screenshots

All screenshots are from the program running in Visual Studio's console window.

Main Menu

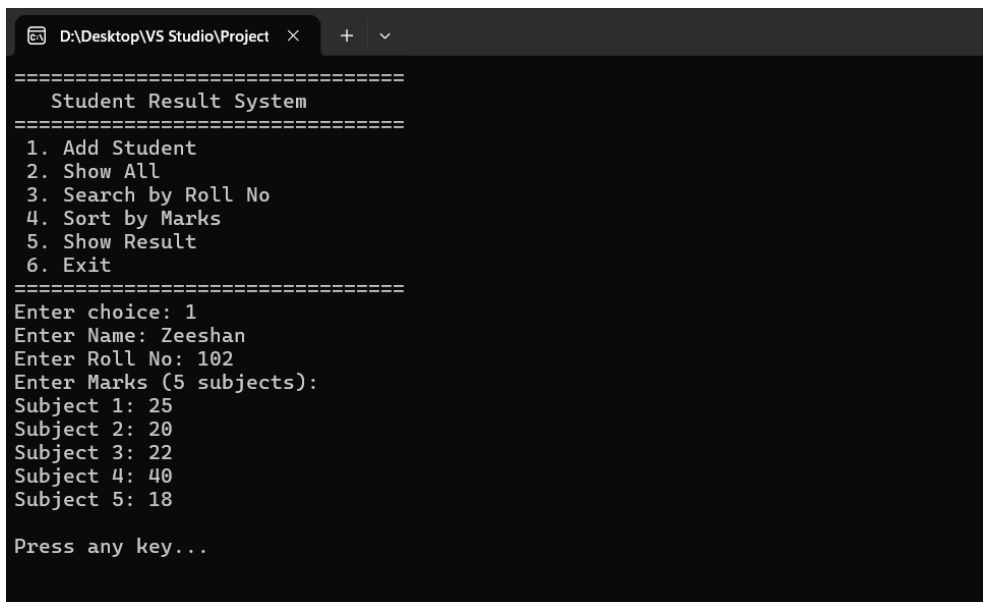


```
D:\Desktop\VS Studio\Project  x  +  v

=====
      Student Result System
=====
1. Add Student
2. Show All
3. Search by Roll No
4. Sort by Marks
5. Show Result
6. Exit
=====
Enter choice: |
```

Figure 1: Main menu on startup

Option 1 — Add Student

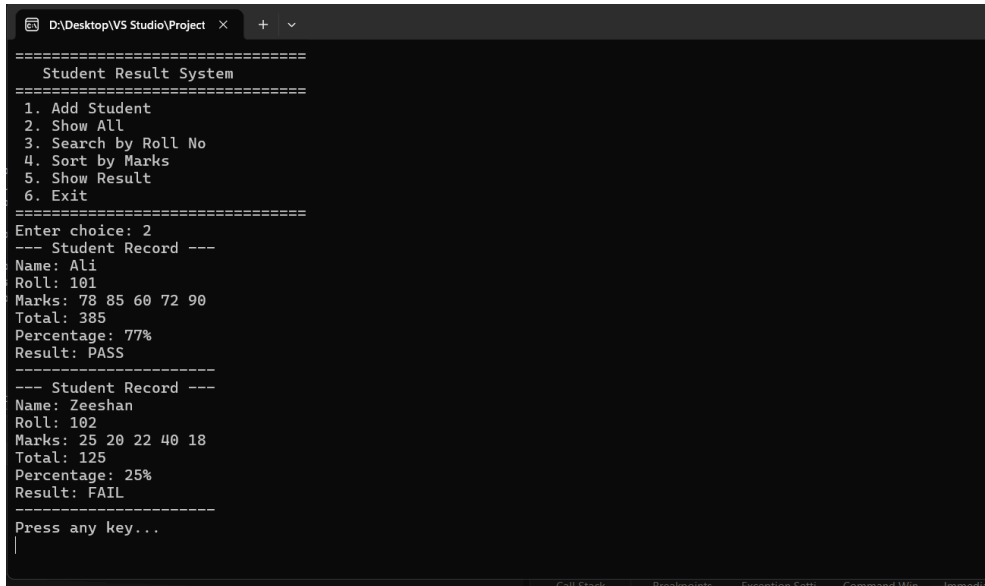


```
D:\Desktop\VS Studio\Project  x  +  v

=====
      Student Result System
=====
1. Add Student
2. Show All
3. Search by Roll No
4. Sort by Marks
5. Show Result
6. Exit
=====
Enter choice: 1
Enter Name: Zeeshan
Enter Roll No: 102
Enter Marks (5 subjects):
Subject 1: 25
Subject 2: 20
Subject 3: 22
Subject 4: 40
Subject 5: 18
Press any key...
```

Figure 2: Adding a student with name, roll number, and 5 subject marks

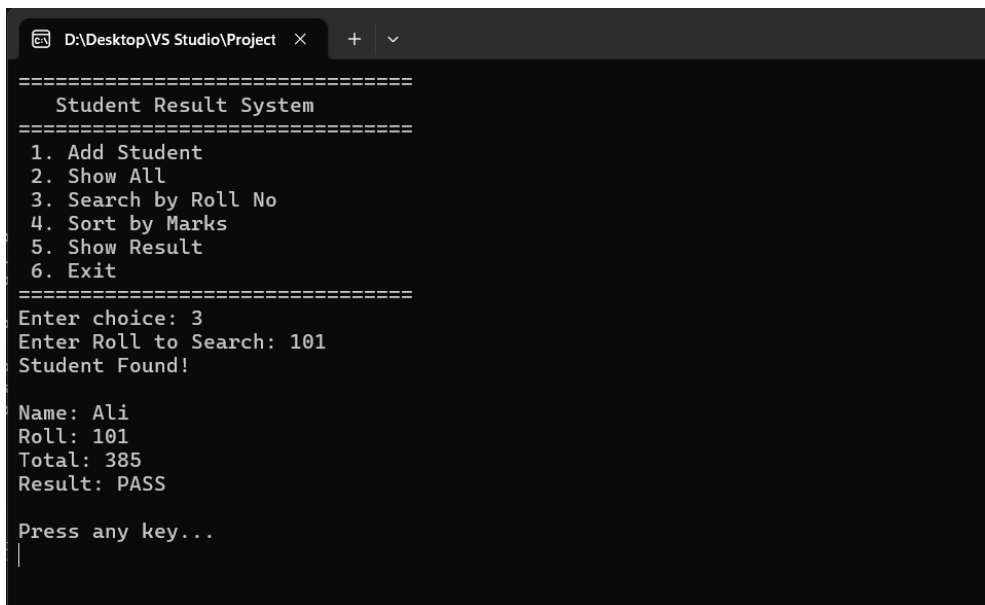
Option 2 — Show All Records



```
=====
Student Result System
=====
1. Add Student
2. Show All
3. Search by Roll No
4. Sort by Marks
5. Show Result
6. Exit
=====
Enter choice: 2
--- Student Record ---
Name: Ali
Roll: 101
Marks: 78 85 60 72 90
Total: 385
Percentage: 77%
Result: PASS
-----
--- Student Record ---
Name: Zeeshan
Roll: 102
Marks: 25 20 22 40 18
Total: 125
Percentage: 25%
Result: FAIL
-----
Press any key...
```

Figure 3: All student records showing marks, totals, percentages, and results

Option 3 — Search by Roll Number



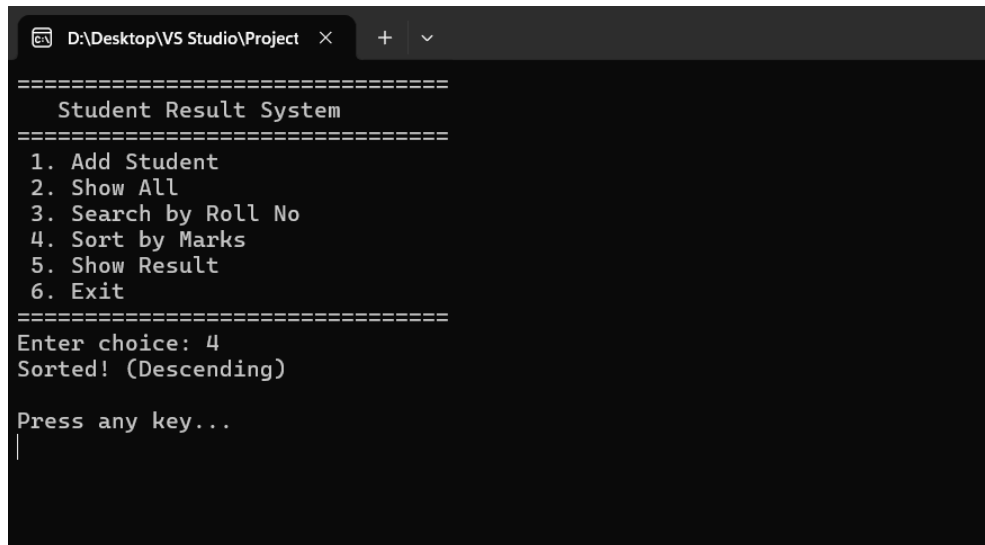
```
=====
Student Result System
=====
1. Add Student
2. Show All
3. Search by Roll No
4. Sort by Marks
5. Show Result
6. Exit
=====
Enter choice: 3
Enter Roll to Search: 101
Student Found!

Name: Ali
Roll: 101
Total: 385
Result: PASS

Press any key...
```

Figure 4: Searching for roll number 101 — student found with result

Option 4 — Sort by Marks

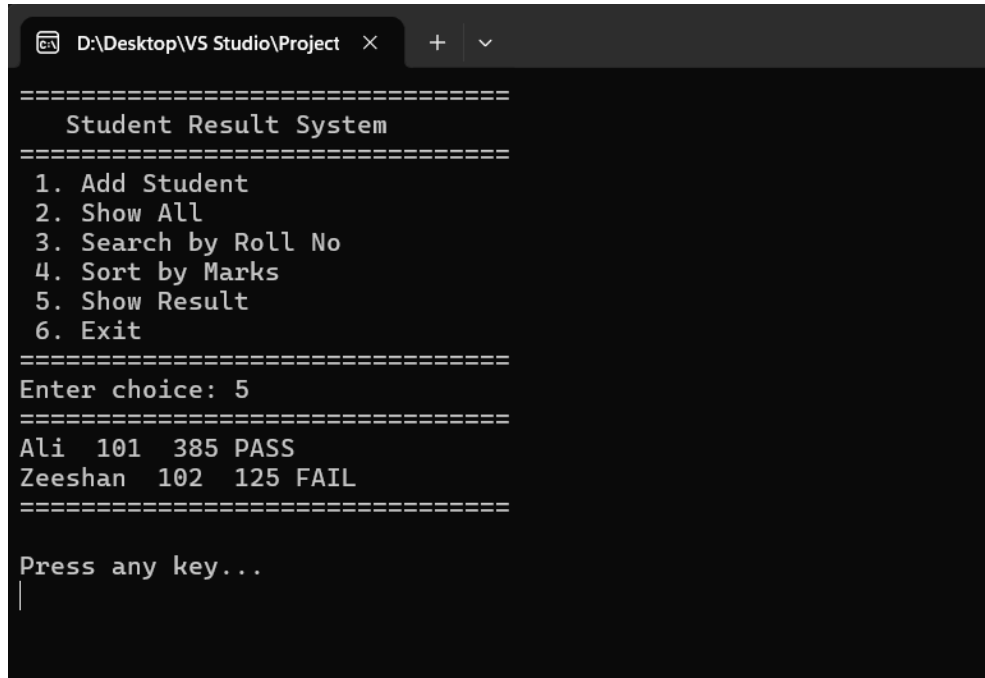


```
D:\Desktop\VS Studio\Project x + v
=====
Student Result System
=====
1. Add Student
2. Show All
3. Search by Roll No
4. Sort by Marks
5. Show Result
6. Exit
=====
Enter choice: 4
Sorted! (Descending)

Press any key...
|
```

Figure 5: Students sorted in descending order by total marks

Option 5 — Result Summary



```
D:\Desktop\VS Studio\Project x + v
=====
Student Result System
=====
1. Add Student
2. Show All
3. Search by Roll No
4. Sort by Marks
5. Show Result
6. Exit
=====
Enter choice: 5
=====
Ali 101 385 PASS
Zeeshan 102 125 FAIL
=====

Press any key...
|
```

Figure 6: Compact result summary showing name, roll, total, and PASS/FAIL

GitHub Repository

We uploaded the source code on GitHub. You can download `main.asm` from there directly. For the full source code (`main.asm`), visit our GitHub repository:

<https://github.com/JunaidAhmed13/StudentResultSystem>

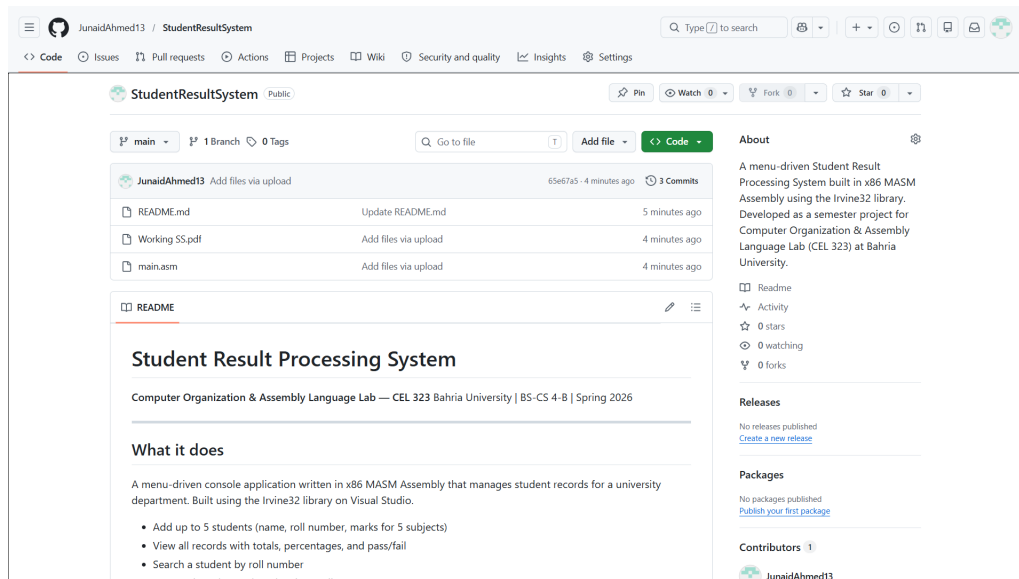


Figure 7: GitHub repository